

CHAPTER 11

an introduction to exotic and path-dependent options



The aim of this Chapter...

... is to give an overview of many of the exciting derivatives above and beyond the basic vanillas. By the end of the chapter you should be able to compare and contrast different sorts of derivative contracts.

In this Chapter...

- how to classify options according to important features
- how to think about derivatives in a way that makes it easy to compare and contrast different contracts
- the names and contract details for many basic types of exotic options

11.1 INTRODUCTION

The contracts we have seen so far are the most basic, and most important, derivative contracts but they only hint at the features that can be found in the more interesting products. Some of these contracts will be explored in following chapters, the more complex ones are covered in *PWOQF2*.

Exotic options are interesting for several reasons. They are harder to price, sometimes being very model dependent. The risks inherent in the contracts are usually more obscure and can lead to unexpected losses. Careful hedging becomes important, whether delta hedging or some form of static hedging to minimize cashflows. Actually, how to hedge exotics is all that really matters. A trader may have a good idea of a reasonable price for an instrument, either from experience or by looking at the prices of similar instruments. But he may not be so sure about the risks in the contract or how to hedge them away successfully.

It is an impossible task to classify all options. The best that we can reasonably achieve is a rough characterization of the most popular of the features to be found in derivative products. I list some of these features in this chapter and give several examples. In the following few chapters I go into more detail in the description of the options and their pricing and hedging. The features that I describe now are time dependence, cashflows, weak path dependence and strong path dependence, dimensionality, the ‘order’ of an option, and finally options with embedded decisions.

One approach that I will not really be taking is to try to decompose an exotic into a portfolio of vanillas. If such a decomposition is exact then the contract was not exotic in the first place. (Riddle: When can you decompose an exotic into a portfolio of vanillas? Answer: When it isn’t exotic.) You may get an idea for an approximate price and how to possibly statically hedge an exotic by such a means but ultimately you will probably have to face a full and proper mathematical analysis of the exotic features in a contract.

11.2 OPTION CLASSIFICATION

In order to figure out how to price and hedge exotic options I have found it incredibly helpful to classify them according to six criteria or features. I can’t overemphasize how useful this has been to me. These six features are unashamedly mathematical in nature, having nothing whatsoever to do with what the name of the option is or what is the underlying. Being mathematical in nature they very quickly give you the following information.

- What kind of pricing method should best be used.
- Whether you can reuse some old code.
- How long it will take you to code it up.
- How fast it will eventually run.

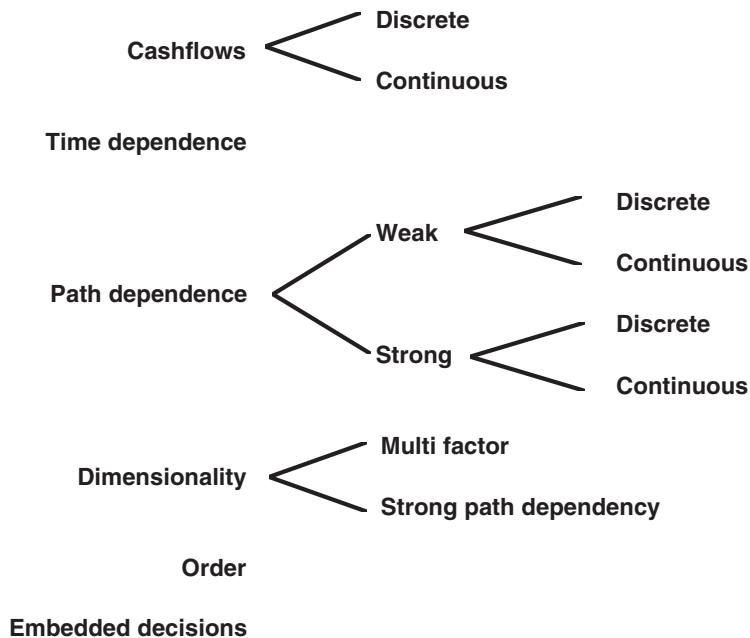


Figure 11.1 Option classification chart.

The six features to look out for are

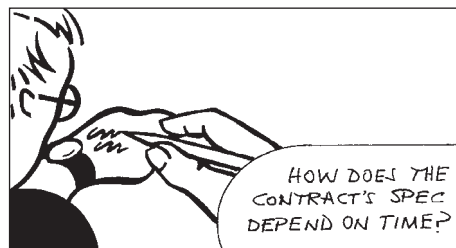
1. Time dependence
2. Cashflows
3. Path dependence
4. Dimensionality
5. Order
6. Embedded decisions

Some of these classes can be broken down further, as shown in Figure 11.1.

Let's now look at these features one by one, in increasing order of interest.

11.3 TIME DEPENDENCE

Time dependence in the present context means that certain specifications within a term sheet vary with time. For example, early exercise might only be permitted on certain dates or during certain periods.



This intermittent early exercise is a characteristic of **Bermudan options**. Such contracts are referred to as time-inhomogeneous.

Time dependence is first on our list of features, since it is not all that earth shattering. Probably the only reason for the importance of time dependence at all is that it requires us to be a little bit careful with any numerical method we employ. Inevitably when solving for an option price via numerical methods we end up needing to do some discretization of time. If the contract has time dependence then we may have restrictions imposed on our discretization.

- Time dependence in an option contract means that our numerical discretization may have to be lined up to coincide with times at, or periods during, which something happens.
- This means that our code will have to keep track of time, dates, etc. This is not difficult, just annoying.



Time Out...

In terms of the binomial model?

If we have time-dependent interest rate or volatility then this just changes the structure of the tree and the discounting. Time dependence in the term sheet means that some nodes might have to be treated differently from others.



11.4 CASHFLOWS

Imagine a contract that pays the holder an amount q at time t_0 . The contract could be a bond and the payment a coupon. If we use $V(t)$ to denote the contract value (ignoring any dependence on any underlying asset) and t_0^- and t_0^+ to denote just before and just after the cashflow date then simple arbitrage considerations lead to

$$V(t_0^-) = V(t_0^+) + q.$$

This is a **jump condition**. The value of the contract jumps by the amount of the cashflow. If this were not the case then there would be an arbitrage opportunity. The behavior of the contract value across the payment date is shown in Figure 11.2.

If the contract is contingent on an underlying variable so that we have $V(S, t)$ then we can accommodate cashflows that depend on the level of the asset, S , i.e. we could have

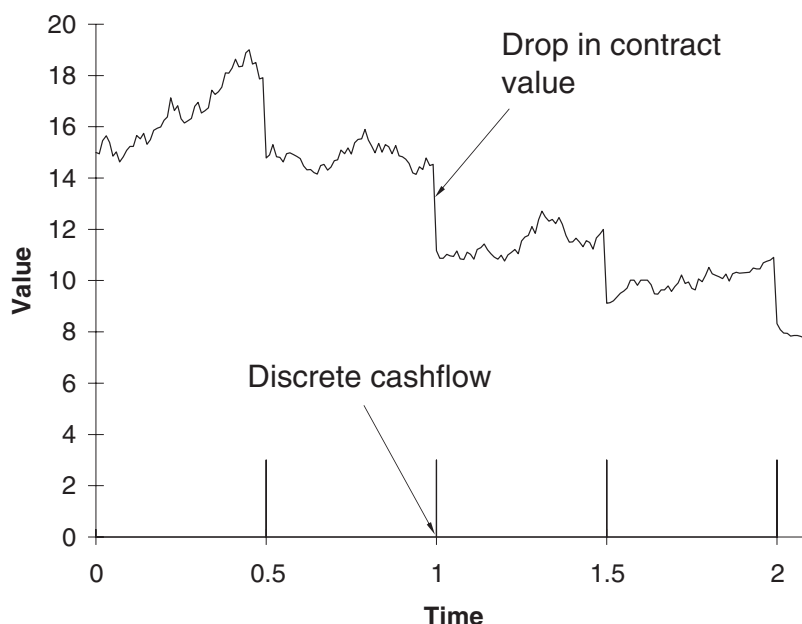


Figure 11.2 A discrete cashflow and its affect on a contract value.

$q(S)$. Furthermore, this also allows us to lump all our options on the same underlying into one large portfolio. Then, across the expiry of each option, there will be a jump in the value of our whole portfolio of the amount of the payoff for that option.

There is one small technical requirement here, the cashflow must be a deterministic function of time and the underlying asset. For example, the contract holder could receive a payment of S^2 , for some asset with price S . However, the above argument would not be valid if, for example, the cashflow depended on the toss of a coin; one dollar is received if heads is thrown and nothing otherwise. The jump condition does not necessarily apply, because the cashflow is not deterministic.

If the cashflow is not deterministic the modeling is not so straightforward. There is no ‘no-arbitrage’ argument to appeal to, and the result could easily depend on an individual’s risk preferences. Nevertheless, we could say, for example, that the jump condition would be that the change in value of the contract would be the *expected* value of the cashflow:

$$V(t_0^-) = V(t_0^+) + E[q].$$

Such a condition would not, however, allow for the risk inherent in the uncertain cashflow.

That was an example of a **discrete cashflow**. You may also see **continuous cashflows** in option contracts. The term sheet may specify something, for example, along the lines of ‘the holder receives \$1 every day that the stock price is below \$80.’ That would effectively be a continuous cashflow. When the cashflow is paid continuously then we no longer have jump conditions, instead we modify the basic Black–Scholes equation to add a source term. We’ll see examples later.

- When a contract has a discretely paid cashflow you should expect to have to apply jump conditions. This also means that the contract has time dependence, see above.

- Continuously paid cashflows mean a modification, although rather simple, to the governing equation.



Time Out...

In terms of the binomial model?

When you work back down the tree you must add in the relevant cashflow to the option value on those nodes where the payment is received.

11.5 PATH DEPENDENCE

Many options have payoffs that depend on the path taken by the underlying asset, and not just the asset's value at expiration. These options are called **path dependent**. Path dependency comes in two varieties, strong and weak.



11.5.1 Strong path dependence

Of particular interest, mathematical and practical, are the strongly path-dependent contracts. These have payoffs that depend on some property of the asset price path in addition to the value of the underlying at the present moment in time; in the equity option language, we cannot write the value as $V(S, t)$. The contract value is a function of at least one more

independent variable. This is best illustrated by an example.

The Asian option has a payoff that depends on the average value of the underlying asset from inception to expiry. We must keep track of more information about the asset price path than simply its present position. The extra information that we need is contained in the 'running average.' This is the average of the asset price from inception until the present, when we are valuing the option. No other information is needed. This running average is then used as a new independent variable, the option value is a function of this as well as the usual underlying and time, and a derivative of the option value with respect to the running average appears in the governing equation. There are many such contracts in existence.

Strong path dependency comes in two forms, **discretely sampled** and **continuously sampled**, depending on whether a discrete subset of asset prices is used or a continuous distribution of them.

- Strong path dependency means that we have to work in higher dimensions. A consequence of this is that our code may take longer to run.

Time Out...

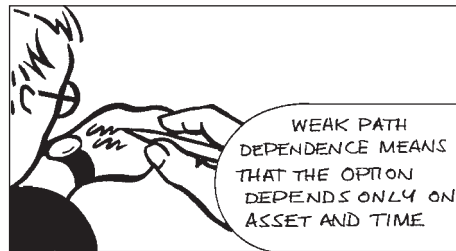
In terms of the binomial model?

This is rather outside the scope of this book. But what you'll have to do is introduce a multi-dimensional tree. There will be branches orthogonal to the S branches to keep track of the path-dependent quantity.



11.5.2 Weak path dependence

A simple example of a contract with weak path dependence is the **barrier**. Barrier (or knock-in, or knock-out) options are triggered by the action of the underlying hitting a prescribed value at some time before expiry. For example, as long as the asset remains below 150, the contract will have a call payoff at expiry. However, should the asset reach this level before expiry then the option becomes worthless; the option has 'knocked out.' This contract is clearly path dependent. Consider the two paths in Figure 11.3; one has a payoff at expiry because the barrier was not triggered, the other is worthless, yet both have the same value of the underlying at expiry.



Time Out...

In terms of the binomial model?

This is usually quite straightforward to implement. But not necessarily very accurate though. For example, with a knock-out barrier option the simplest way to price this is to set all option values to zero beyond the barrier.



We shall see in Chapter 13 that such a contract is only weakly path dependent: we still solve a partial differential equation in the two variables, the underlying and time.

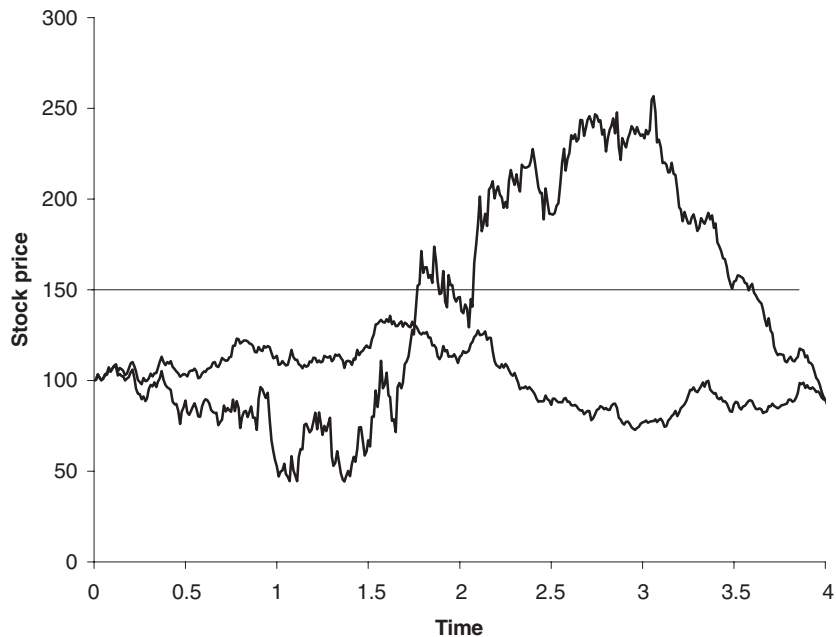
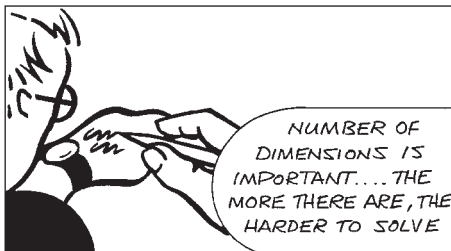


Figure 11.3 Two paths, having the same value at expiry, but with completely different payoffs.

And that is the difference, mathematically speaking, between strong and weak path dependency. A weakly path-dependent contract does not require us to introduce an extra variable to handle the path dependency. Again, we can imagine discrete and continuous versions.

- Weak path dependency means that we *don't* have to work in higher dimensions, so our code should be pretty fast.



11.6 DIMENSIONALITY

Dimensionality refers to the number of underlying independent variables. The vanilla option has two independent variables, S and t , and is thus two dimensional. The weakly path-dependent contracts have the same number of dimensions as their non-path-dependent cousins, i.e. a barrier call option has the same two dimensions as a vanilla call. The roles

of the asset dimension and the time dimension are quite different from each other, as discussed in Chapter 7 on the diffusion equation. This is because the governing equation, the Black–Scholes equation, contains a second asset-price derivative but only a first time derivative.

We can have two types of three-dimensional problem. The first type of problem that is three dimensional is the strongly path-dependent contract. Typically, the new independent variable is a measure of the path-dependent quantity on which the option is contingent. The new variable may be the average of the asset price to date, say. In this case, derivatives of the option value with respect to this new variable are only of the first order. Thus the new variable acts more like another time-like variable. This topic is covered in detail in *PWOQF2*.

The second type of three-dimensional problem occurs when we have a second source of randomness, such as a second underlying asset. We might, for example, have an option on the maximum of two equities. Both of these underlyings are stochastic, each with a volatility, and there will be a correlation between them. In the governing equation we will see a second derivative of the option value with respect to each asset. We say that there is diffusion in both S_1 and S_2 .

- Higher dimensions means longer computing time.
- The number of dimensions we have also tells us what kind of numerical method to use. High dimensions mean that we probably want to use Monte Carlo, low means finite difference.

Time Out...

In terms of the binomial model?

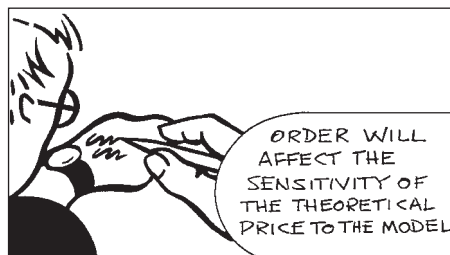
As with path dependency you will have to have a higher-dimensional tree structure to model the new dependent variables.



11.7 THE ORDER OF AN OPTION

The next classification that we make is the **order** of an option. Not only is this a classification but the idea also introduces fundamental modeling issues.

The basic vanilla options are of first order. Their payoffs depend only on the underlying asset, the quantity that we are *directly* modeling. Other, path-dependent, contracts can still be of first order if the payoff depends only on properties of the asset price path. 'Higher order' refers to options whose payoff, and hence value, is contingent on the value of *another* option. The obvious



second-order options are compound options, for example a call option giving the holder the right to buy a put option. The compound option expires at some date T_1 and the option on which it is contingent, expires at a later time T_2 . Technically speaking, such an option is weakly path dependent. The *theoretical* pricing of such a contract is straightforward as we shall see.

From a practical point of view, the compound option raises some important modeling issues: the payoff for the compound option depends on the *market* value of the underlying option, and not on the theoretical price. If you hold a compound option and want to exercise the first option then you must take possession of the underlying option. If that option is worth less than you think it should (because your model says so) then there is not much you can do about it. High-order option values are very sensitive to the basic pricing model and should be handled with care.

- When an option is second or higher order we have to solve for the first-order option, first. We thus have a layer cake, we must work on the lower levels and the results of those feed into the higher levels.
- This means that computationally we have to solve more than one problem to price our option.



Time Out...

In terms of the binomial model?

You will need one tree structure to model the 'underlying option' and this will feed into the tree for the higher-order option.

11.8 EMBEDDED DECISIONS

We have seen early exercise in the American option problem. Early exercise is a common feature of other contracts, perhaps going by other names. For example, the conversion of convertible bonds is mathematically identical to the early exercise of an American option. The key point about early exercise is that the holder of this valuable right should ideally act *optimally*, i.e. they must decide *when* to exercise or convert. In the partial differential equation framework that has been set up, this optimality is achieved by solving a free boundary problem, with a constraint on the option value, together with a smoothness condition. It is this smoothness condition, that the derivative of the option value with respect to the underlying is continuous, that ensures optimality, i.e. maximization of the

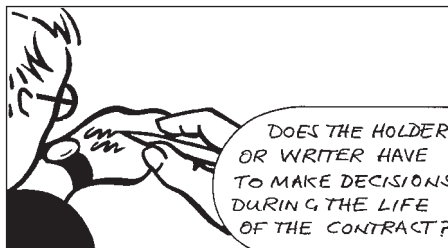
option value with respect to the exercise or conversion strategy. It is perfectly possible for there to be more than one early-exercise region.¹

Holding an American option you are faced with the decision whether and when to exercise your rights. The American option is the most common contract that contains within it a decision feature. Other contracts require more subtle and interesting decisions to be made.

When a contract has embedded decisions you need an algorithm for deciding how that decision will be made. That algorithm amounts to assuming that the holder of the contract acts to *make the option value as high as possible for the delta-hedging writer*. The pricing algorithm then amounts to searching across all possible holder decision strategies for the one that maximizes the option value. That sounds hard, but approached correctly is actually remarkably straightforward, especially if you use the finite-difference method. The justification for seeking the strategy that maximizes the value is that the writer cannot afford to sell the option for anything less, otherwise he would be exposed to ‘decision risk.’

When the option writer or issuer is the one with the decision to make then the value is based on seeking the strategy that minimizes the value.

- Decision features mean that we’d really like to price via finite differences.
- The code will contain a line in which we seek the best price, so watch out for $\geq$ or $\leq$ signs.



Time Out...

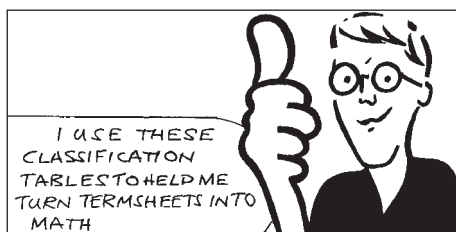
In terms of the binomial model?

You’ve seen this already in Chapter 3. Just make sure that the option value stays above the payoff at all times that exercise is allowed.

For other contracts, whether this is easy or not depends on the nature of the contract. But as with the American option, decision making requires a process for ensuring optimality. Mathematically this usually boils down to a $\geq$ sign somewhere.



¹ One rarely mentioned aspect of American options, and, generally speaking, contracts with early exercise-type characteristics, is that they are path dependent. Whether the owner of the option still holds the option at expiry depends on whether or not he has exercised the option, and thus on the path taken by the underlying. For American-type options this path dependence is weak, in the sense that the partial differential equation to be solved has no more independent variables than a similar, but European, contract.



11.9 CLASSIFICATION TABLES

I find tables like the following very useful for the classification of special contracts.

Classification	Option Name
Time dependence	Do details vary with time? Eg. discrete sampling.
Cashflow	Does money change hands during life of contract?
Path dependence	Weak or Strong?
Dimension	2, 3, 4, ... ?
Order	First, second, ... ?
Decisions	Does holder and/or writer have to make decisions?

11.10 EXAMPLES OF EXOTIC OPTIONS

There now follow some basic examples, just to get you into the swing of things.

11.10.1 Compounds and choosers

Compound and **chooser options** are simply options on options. The compound option gives the holder the right to buy (call) or sell (put) another option. Thus we can imagine owning a call on a put, for example. This gives us the right to buy a put option for a specified amount on a specified date. If we exercise the option then we will own a put option which gives us the right to sell the underlying. This compound option is second order because the compound option gives us rights over another derivative. Although the Black–Scholes model can theoretically cope with second-order contracts it is not so clear that the model is completely satisfactory in practice; when we exercise the contract we get an option at the market price, not at our theoretical price.

In the Black–Scholes framework the compound option is priced as follows. There are two steps: first price the underlying option and then price the compound option. Suppose that the underlying option has a payoff of $F(S)$ at time T , and that the compound option can be exercised at time $T_{Co} < T$ to get $G(V(S, T_{Co}))$ where $V(S, t)$ is the value of the underlying option. Step one is to price the underlying option, i.e. to find $V(S, t)$. This satisfies

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0 \quad \text{with} \quad V(S, T) = F(S).$$

Solve this problem so that you have found $V(S, T_{Co})$. This is the (theoretical) value of the underlying option at time T_{Co} , which is the time at which you can exercise your compound option. Now comes the second step, to value the compound option. The value of this is $Co(S, t)$ which satisfies

$$\frac{\partial Co}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 Co}{\partial S^2} + rS \frac{\partial Co}{\partial S} - r Co = 0 \quad \text{with} \quad Co(S, T_{Co}) = G(V(S, T_{Co})).$$

As an example, if we have a call on a call with exercise prices E for the underlying and E_{Co} for the compound option, then we have

$$F(S) = \max(S - E, 0) \quad \text{and} \quad G(V) = \max(V - E_{Co}, 0).$$

In Figure 11.4 is shown the value of a vanilla call option at the time of expiry of a put option on this call. This is obviously some time before the expiry of the underlying call. In the same figure is the payoff for the put on this option. This is the final condition for the Black–Scholes partial differential equation.

It is possible to find analytical formulæ for the price of basic compound options in the Black–Scholes framework when volatility is constant. These formulæ involve the cumulative distribution function for a bivariate Normal variable. However, because of the second-order nature of compound options and thus their sensitivity to the precise nature of the asset price random walk, these formulæ are dangerous to use in practice. Practitioners use either a stochastic volatility model or an implied volatility surface.

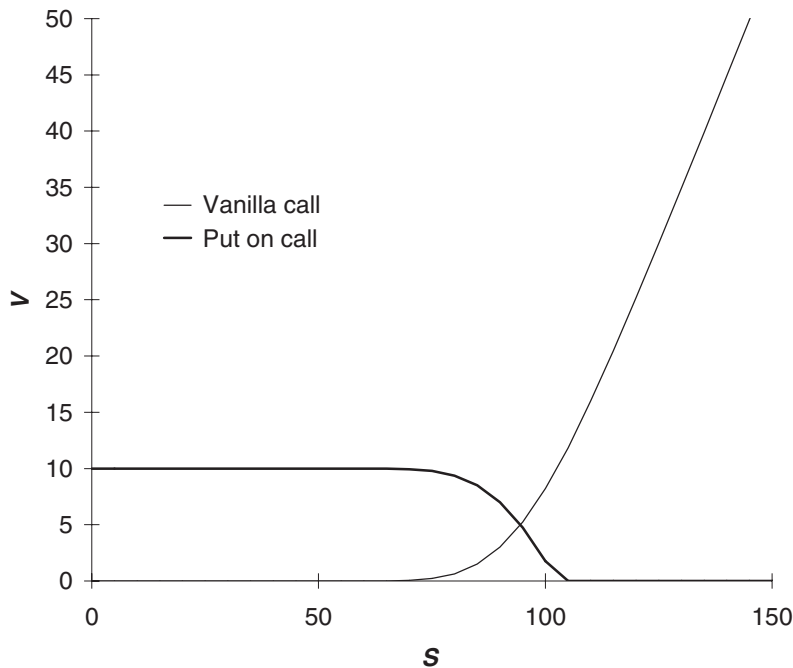


Figure 11.4 The value of a vanilla call option some time before expiry and the payoff for a put on this option.

Chooser options are similar to compounds in that they give the holder the right to buy a further option. With the chooser option the holder can choose whether to receive a call or a put, for example. Generally, we can write the value of the chooser option as $Ch(S, t)$ and the value of the underlying options as $V_1(S, t)$ and $V_2(S, t)$ (or more). Now

$$\frac{\partial Ch}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 Ch}{\partial S^2} + rS \frac{\partial Ch}{\partial S} - rCh = 0,$$

$$\frac{\partial V_1}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V_1}{\partial S^2} + rS \frac{\partial V_1}{\partial S} - rV_1 = 0$$

and

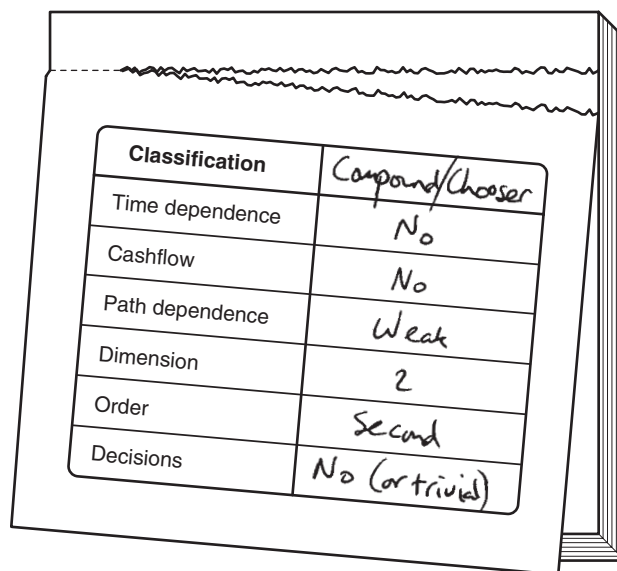
$$\frac{\partial V_2}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V_2}{\partial S^2} + rS \frac{\partial V_2}{\partial S} - rV_2 = 0.$$

Final conditions are the usual payoffs for the underlying options at their expiry dates and

$$Ch(S, T_{Ch}) = \max(V_1(S, T_{Ch}) - E_1, V_2(S, T_{Ch}) - E_2, 0),$$

with the obvious notation.

The practical problems with pricing choosers are the same as for compounds.



Classification	Compound/Chooser
Time dependence	No
Cashflow	No
Path dependence	Weak
Dimension	2
Order	Second
Decisions	No (or trivial)

In Figure 11.5 is shown the values of a vanilla call and a vanilla put some time before expiry. In the same figure is the payoff for a call on the best of these two options (less an exercise price). This is the final condition for the Black–Scholes partial differential equation.

Figures 11.6 and 11.7 show the Bloomberg screens for valuing chooser options.

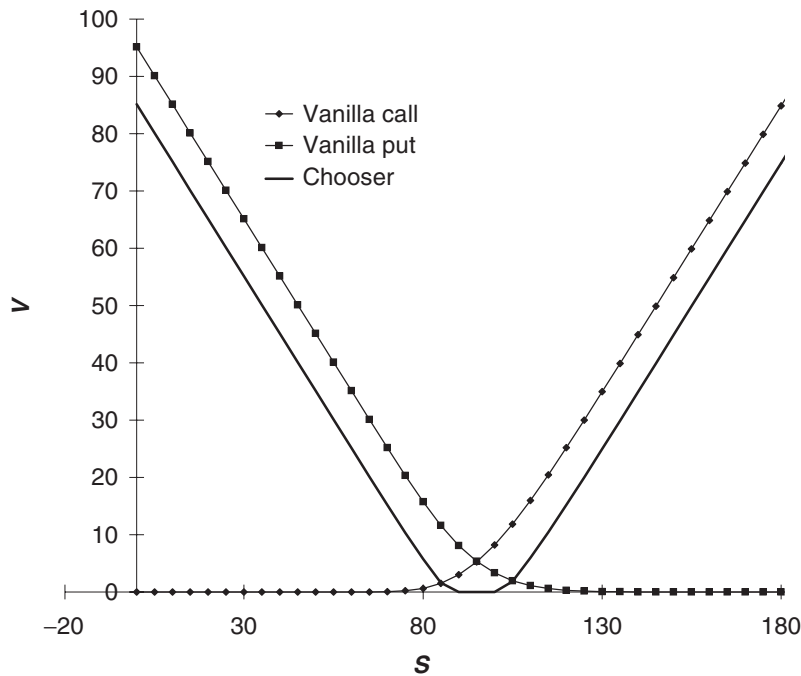


Figure 11.5 The value of a vanilla call option and a vanilla put option some time before expiry and the payoff for the best of these two.

<HELP> for explanation.		DL18 Index OVX	
Chooser Option Valuation		Page 1 of 2	
DJZ9	DJIA INDEX FUT	Dec99	Currency: USD
Price of DJZ9 Index	11150	Hit 1 GO for save/send screen Hit 2 GO for notes Hit 3 GO to price off futrd, divds Hit MENU for exotic option types Hit PAGE for scenario graph	
Call Strike:	11150.00	(USD)Rate:	4.767% Semiannual
Put Strike:	11150.00		
Trade Date:	9/11/99	Choice Date:	10/29/99
Settlement Date:	9/11/99	Exercise Type of chosen option:	E European
Days to Exp. of Call:	96	Days to Exp. of Put:	96
Expir. Date of Call:	12/16/99	Expir. Date of Put:	12/16/99
Settle Date of Call:	10/29/99	Settle Date of Put:	10/29/99
Exercise Delay of Chosen Option:	0		
Option Valuation and Risk Parameters			Dividends
Value	Percent	Time Value:	571.19519
Price: 571.195	5.123%	7-Day Decay:	29.86389
Volatility: 14.853%		Premium:	5.12283
Delta: 0.08757		Parity:	0.00000
Gamma: 0.00112		Gearing:	19.52047
Vega: 38.44012			
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Figure 11.6 Bloomberg chooser option valuation screen. Source: Bloomberg L.P.

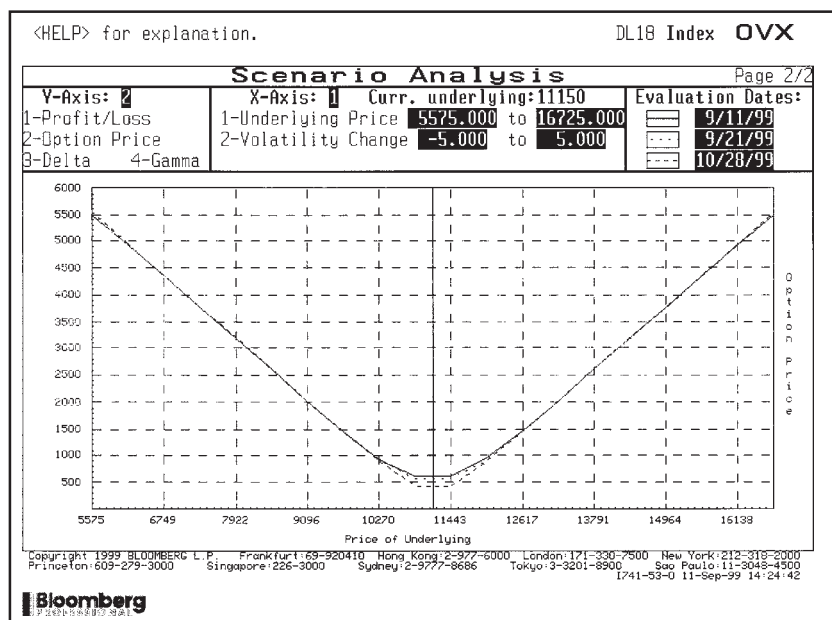
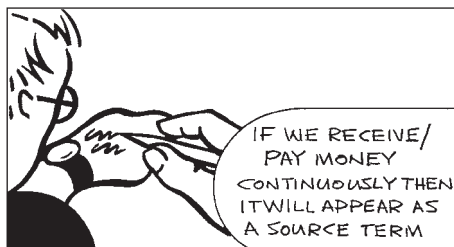


Figure 11.7 Bloomberg scenario analysis for a chooser. Source: Bloomberg L.P.

Extendible options are very, very similar to compounds and choosers. At some specified time the holder can choose to accept the payoff for the original option or to extend the option's life and even change the strike. Sometimes it is the writer who has these powers of extension. The reader has sufficient knowledge to be able to model these contracts in the Black–Scholes framework.

11.10.2 Range notes

Range notes are very popular contracts, existing on the 'lognormal' assets such as equities and currencies, and as fixed-income products. In its basic, equity derivative, form the range note pays at a rate of L all the time that the underlying lies within a given range, $S_l \leq S \leq S_u$. That is, for every dt that the asset is in the range you receive $L dt$. Introducing $I(S)$ as the function taking the value 1 when $S_l \leq S \leq S_u$ and zero otherwise, the range note satisfies



$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV + LI(S) = 0.$$

In Figure 11.8 is shown the term sheet for a range note on the Mexican peso, US dollar exchange rate. This contract pays out the positive part of the difference between number of days the exchange rate is inside the range less the number of days outside the range. This payment is received at expiry. (This contract is subtly different, and more complicated than the basic range note described above. Why?)

Classification	Range Note
Time dependence	No
Cashflow	Yes (continuous)
Path dependence	Weak
Dimension	2
Order	first
Decisions	No

*Preliminary and Indicative
For Discussion Purposes Only*

6 Month In-Out Range Accrual Option on MXN/USD FX Rate

Settlement Date	One week from Trade Date
Maturity Date	6 months from Trade Date
Option Premium	USD 50,000+
Option Type	In MINUS Out Range Accrual on MXN/USD FX rate
Option Payment Date	2 business days after Maturity Date
Option Payout	USD 125,000 * Index
Where Index	$\frac{\text{FX daily In MINUS FX daily Out}}{\text{Total Business Days}} \text{ (subject to a minimum of zero)}$
FX daily In	The number of business days Spot MXN/USD Exchange Rate is within Range
FX daily Out	The number of business days Spot MXN/USD Exchange Rate is outside Range
Range	MXN/USD 7.7200-8.1300
Spot MXN/USD Exchange Rate	Official spot exchange rate as determined by the Bank of Mexico as appearing on Reuters page "BNMX" at approximately 3:00 p.m. New York time.
Current Spot MXN/USD	7.7800

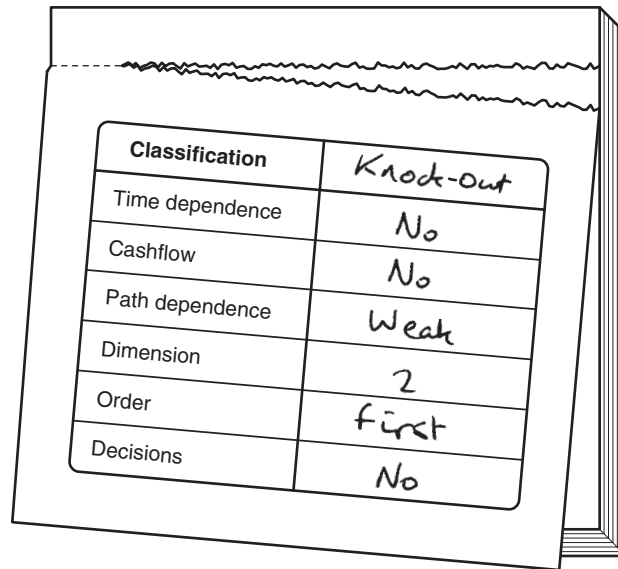
This indicative termsheet is neither an offer to buy or sell securities or an OTC derivative product which includes options, swaps, forwards and structured notes having similar features to OTC derivative transactions, nor a solicitation to buy or sell securities or an OTC derivative product. The proposal contained in the foregoing is not a complete description of the terms of a particular transaction and is subject to change without limitation.

Figure 11.8 Term sheet for an in-out range accrual note on MXN/USD.

11.10.3 Barrier options

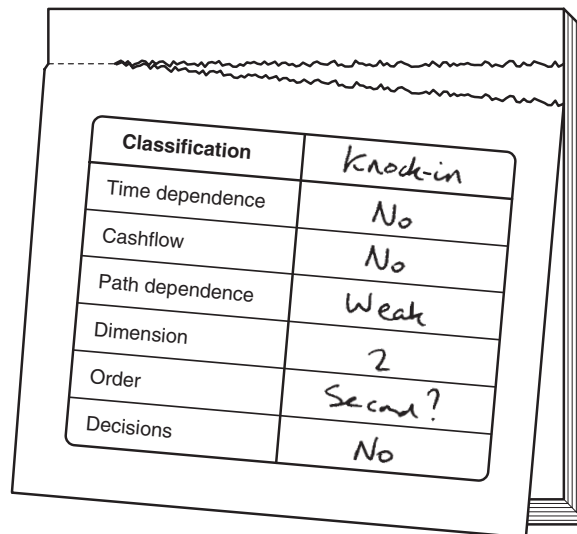
Barrier options have a payoff that is contingent on the underlying asset reaching some specified level before expiry. The critical level is called the barrier, there may be more than one. Barrier options are weakly path dependent. Barrier options are discussed in depth in Chapter 13.

Barrier options come in two main varieties, the 'in' barrier option (or **knock-in**) and the 'out' barrier option (or **knock-out**). The former only have a payoff if the barrier level is reached before expiry and the latter only have a payoff if the barrier is *not* reached before



A hand-drawn table on a piece of paper with a torn top edge, tilted slightly to the right. The table has two columns: 'Classification' and 'Knock-out'. It lists seven characteristics and their corresponding values for a knock-out option.

Classification	Knock-out
Time dependence	No
Cashflow	No
Path dependence	Weak
Dimension	2
Order	first
Decisions	No



A hand-drawn table on a piece of paper with a torn top edge, tilted slightly to the right. The table has two columns: 'Classification' and 'Knock-in'. It lists seven characteristics and their corresponding values for a knock-in option.

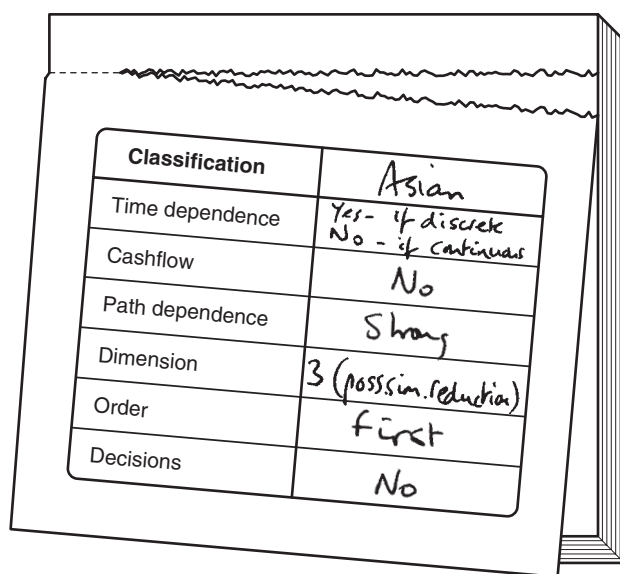
Classification	Knock-in
Time dependence	No
Cashflow	No
Path dependence	Weak
Dimension	2
Order	Second?
Decisions	No

expiry. These contracts are weakly path dependent, meaning that the price depends only on the current level of the asset and the time to expiry. They satisfy the Black–Scholes equation, with special boundary conditions as we shall see.

11.10.4 Asian options

Asian options have a payoff that depends on the average value of the underlying asset over some period before expiry. They are the first strongly path-dependent contract we examine. They are strongly path dependent because their value prior to expiry depends on the path taken and not just on where they have reached. Their value depends on the *average to date* of the asset. This average to date will be very important to us, we introduce something like it as a new state variable. (In *PWOQF2* it is shown how to derive a partial differential equation for the value of this Asian contract, but now the differential equation will have *three* independent variables.)

The average used in the calculation of the option's payoff can be defined in many different ways. It can be an arithmetic average or a geometric average, for example. The data could be continuously sampled, so that every realized asset price over the given period is used. More commonly, for practical and legal reasons, the data is usually sampled discretely; the calculated average may only use every Friday's closing price, for example.



Classification	Asian
Time dependence	Yes - if discrete No - if continuous
Cashflow	No
Path dependence	Strong
Dimension	3 (poss. sim. reduction)
Order	first
Decisions	No

11.10.5 Lookback options

Lookback options have a payoff that depends on the realized maximum or minimum of the underlying asset over some period prior to expiry. An extreme example, which captures the flavor of these contracts, is the option that pays off the difference between that maximum realized value of the asset and the minimum value over the next year. Thus

it enables the holder to buy at the lowest price and sell at the highest, every trader's dream. Of course, this payoff comes at a price. And for such a contract that price would be very high.

Again the maximum or minimum can be calculated continuously or discretely, using every realized asset price or just a subset. In practice the maximum or minimum is measured discretely.

Classification	Lookback
Time dependence	Yes - if discrete No - if continuous
Cashflow	No
Path dependence	Strong
Dimension	3 (poss. sim. reduction)
Order	first
Decisions	No

||.|| SUMMARY OF MATH/CODING CONSEQUENCES

Classification	Examples	Consequences
Time dependence	Bermudan exercise, discrete sampling, ...	Must keep track of time in code
Cashflow	Swap, instalments, ...	Jump in option value/Source term in pde
Path dependence	Barrier, Asian, lookback, ...	If strong path dependency need extra dimension
Dimension	Strongly path dependent, multi asset, ...	Monte Carlo may be better than finite difference
Order	Compounds, in barriers, ...	Solve lower-level option(s) first and input into higher
Decisions	American, passport, chooser, ...	Finite difference better than Monte Carlo, 'optimize'

11.12 SUMMARY

This chapter suggests ways to think about derivative contracts that make their analysis simpler. To be able to make comparisons between different contracts is a big step forward in understanding them.

FURTHER READING

- Geske (1979) discusses the valuation of compound options.
- See Taleb (1997) for more details of classifications of the type I have described. This book is an excellent and entertaining read.
- The book by Zhang (1997) is a discussion of many types of exotic options, with many formulæ.
- See Kyprianou *et al.* (2005) for Lévy processes and exotic option pricing.
- *PWOQF2* has many chapters on exotics and the mathematics used for pricing them.

SOME FORMULÆ FOR ASIAN OPTIONS

There are very few nice formulæ for the values of Asian options. The most well known are for average rate calls and puts when the average is a continuously sampled, geometrical average.

The geometric average rate call This option has payoff

$$\max(A - E, 0),$$

where A is the continuously sampled geometric average. This option has a Black–Scholes value of

$$e^{-r(T-t)} \left(G \exp \left(\frac{(r - D - \sigma^2/2)(T-t)^2}{2T} + \frac{\sigma^2(T-t)^3}{6T^2} \right) N(d_1) - EN(d_2) \right)$$

where

$$I = \int_0^t \log(S(\tau)) d\tau,$$

$$G = e^{I/T} S^{(T-t)/T},$$

$$d_1 = \frac{T \log(G/E) + (r - D - \sigma^2/2)(T-t)^2/2 + \sigma^2(T-t)^3/3T}{\sigma \sqrt{(T-t)^3/3}}$$

and

$$d_2 = \frac{T \log(G/E) + (r - D - \sigma^2/2)(T-t)^2/2}{\sigma \sqrt{(T-t)^3/3}}.$$

The geometric average of a lognormal random walk is itself lognormally distributed, but with a reduced volatility.

The geometric average rate put This option has payoff

$$\max(E - A, 0),$$

where A is the continuously sampled geometric average. This option has a Black–Scholes value of

$$e^{-r(T-t)} \left(EN(-d_2) - G \exp \left(\frac{(r-D-\sigma^2/2)(T-t)^2}{2T} + \frac{\sigma^2(T-t)^3}{6T^2} \right) N(d_1) \right).$$

SOME FORMULÆ FOR LOOKBACK OPTIONS

Floating strike lookback call The continuously sampled version of this option has a payoff

$$\max(S - M, 0) = S - M,$$

where M is the realized minimum of the asset price. In the Black–Scholes world the value is

$$Se^{-D(T-t)}N(d_1) - Me^{-r(T-t)}N(d_2) + Se^{-r(T-t)} \frac{\sigma^2}{2(r-D)} \left(\left(\frac{S}{M} \right)^{-\frac{2(r-D)}{\sigma^2}} N \left(-d_1 + \frac{2(r-D)\sqrt{T-t}}{\sigma} \right) - e^{(r-D)(T-t)} N(-d_1) \right),$$

where

$$d_1 = \frac{\log(S/M) + (r-D + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}$$

and

$$d_2 = d_1 - \sigma\sqrt{T-t}.$$

Floating strike lookback put The continuously sampled version of this option has a payoff

$$\max(M - S, 0) = M - S,$$

where M is the realized maximum of the asset price. The value is

$$Me^{-r(T-t)}N(-d_2) - Se^{-D(T-t)}N(-d_1) + Se^{-r(T-t)} \frac{\sigma^2}{2(r-D)} \left(- \left(\frac{S}{M} \right)^{-\frac{2(r-D)}{\sigma^2}} N \left(d_1 - \frac{2(r-D)\sqrt{T-t}}{\sigma} \right) + e^{(r-D)(T-t)} N(d_1) \right),$$

where

$$d_1 = \frac{\log(S/M) + (r-D + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}$$

and

$$d_2 = d_1 - \sigma\sqrt{T-t}.$$

Fixed strike lookback call This option has a payoff given by

$$\max(M - E, 0)$$

where M is the realized maximum. For $E > M$ the fair value is

$$\begin{aligned} & Se^{-D(T-t)}N(d_1) - Ee^{-r(T-t)}N(d_2) \\ & + Se^{-r(T-t)}\frac{\sigma^2}{2(r-D)}\left(-\left(\frac{S}{E}\right)^{-\frac{2(r-D)}{\sigma^2}}N\left(d_1 - \frac{2(r-D)\sqrt{T-t}}{\sigma}\right) + e^{(r-D)(T-t)}N(d_1)\right), \end{aligned}$$

where

$$d_1 = \frac{\log(S/E) + (r - D + \frac{1}{2}\sigma^2)(T - t)}{\sigma\sqrt{T-t}}$$

and

$$d_2 = d_1 - \sigma\sqrt{T-t}.$$

When $E < M$ the value is

$$\begin{aligned} & (M - E)e^{-r(T-t)} + Se^{-D(T-t)}N(d_1) - Me^{-r(T-t)}N(d_2) \\ & + Se^{-r(T-t)}\frac{\sigma^2}{2(r-D)}\left(-\left(\frac{S}{M}\right)^{-\frac{2(r-D)}{\sigma^2}}N\left(d_1 - \frac{2(r-D)\sqrt{T-t}}{\sigma}\right) + e^{(r-D)(T-t)}N(d_1)\right), \end{aligned}$$

where

$$d_1 = \frac{\log(S/M) + (r - D + \frac{1}{2}\sigma^2)(T - t)}{\sigma\sqrt{T-t}}$$

and

$$d_2 = d_1 - \sigma\sqrt{T-t}.$$

Fixed strike lookback put This option has a payoff given by

$$\max(E - M, 0)$$

where M is the realized minimum. For $E < M$ the fair value is

$$\begin{aligned} & Ee^{-r(T-t)}N(-d_2) - Se^{-D(T-t)}N(-d_1) \\ & + Se^{-r(T-t)}\frac{\sigma^2}{2(r-D)}\left(\left(\frac{S}{E}\right)^{-\frac{2(r-D)}{\sigma^2}}N\left(-d_1 + \frac{2(r-D)\sqrt{T-t}}{\sigma}\right) - e^{(r-D)(T-t)}N(-d_1)\right), \end{aligned}$$

where

$$d_1 = \frac{\log(S/E) + (r - D + \frac{1}{2}\sigma^2)(T - t)}{\sigma\sqrt{T-t}}$$

and

$$d_2 = d_1 - \sigma\sqrt{T-t}.$$

When $E > M$ the value is

$$(E - M)e^{-r(T-t)} - Se^{-D(T-t)}N(-d_1) + Me^{-r(T-t)}N(-d_2) \\ + Se^{-r(T-t)}\frac{\sigma^2}{2(r-D)}\left(\left(\frac{S}{M}\right)^{-\frac{2(r-D)}{\sigma^2}}N\left(-d_1 + \frac{2(r-D)\sqrt{T-t}}{\sigma}\right) - e^{(r-D)(T-t)}N(-d_1)\right),$$

where

$$d_1 = \frac{\log(S/M) + (r - D + \frac{1}{2}\sigma^2)(T - t)}{\sigma\sqrt{T - t}}$$

and

$$d_2 = d_1 - \sigma\sqrt{T - t}.$$

EXERCISES

1. A chooser option has the following properties:

At time $T_C < T$, the option gives the holder the right to buy a European call or put option with exercise price E and expiry at time T , for an amount E_C . What is the value of this option when $E_C = 0$?

Hint: Write down the payoff of the option and then use put-call parity to simplify the result.

2. How would we value the chooser option in the above question if E_C was non-zero?
3. Prove put-call parity for European compound options:

$$C_C + P_P - C_P - P_C = S - E_2e^{-r(T_2-t)},$$

where C_C is a call on a call, C_P is a call on a put, P_C is a put on a call and P_P is a put on a put. The compound options have exercise price E_1 and expiry at time T_1 and the underlying calls and puts have exercise price E_2 and expiry at time T_2 .

4. Find the value of the power European call option. This is an option with exercise price E , expiry at time T , when it has a payoff:

$$\Lambda(S) = \max(S^2 - E, 0).$$

Hint: Note that if the underlying asset price is assumed to be lognormally distributed then the square of the price is also lognormally distributed.